# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## CLASS 6 — SCIENCE · HARD LEVEL UNIT TEST SET 2

Chapters 5–8: **Measurement of Length and Motion, Materials Around Us, Temperature and its Measurement, A Journey through States of Water** | RBSE Pattern | Session 2026–27

**Time: 2 Hours** Name: _____ Roll No.: _____ **Maximum Marks: 50**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 15 × 1 = 15

*Choose the correct option and write it in your answer book.*

**1.** A student heats 100 g of ice and records temperatures. The graph has a flat portion at 0 °C and another at 100 °C. The flat portions represent: 1

**(a)** cooling **(b)** freezing
**(c)** melting and boiling **(d)** heating of solid and liquid

**2.** Which of the following is true about a freely falling stone? 1

**(a)** it moves in a circle
**(b)** it is at rest
**(c)** it is in linear motion and its speed increases
**(d)** it is in periodic motion

**3.** If material C is an opaque metal and D is opaque wood, how would you distinguish them? 1

**(a)** C conducts heat and D does not **(b)** both float
**(c)** both transparent **(d)** both dissolve

**4.** If a 20 °C object touches a 60 °C object, which gets warmer? 1

**(a)** the 60 °C object **(b)** the 20 °C object **(c)** both cool **(d)** neither

**5.** If 50 mL of water evaporates from a vessel in 5 days, the average loss per day is: 1

**(a)** 50 mL **(b)** 250 mL **(c)** 5 mL **(d)** 10 mL

**6.** A student measures a pencil three times and gets 8.3 cm, 8.2 cm and 8.4 cm. The best value to report is: 1

**(a)** 8.3 cm **(b)** 8.2 cm **(c)** 8.4 cm **(d)** 24.9 cm

**7.** Which of the following is a reason to use thermocol for packing glass items? 1

**(a)** it is transparent **(b)** it is a metal
**(c)** it is light and cushions shocks **(d)** it is hard

**8.** A student heats a metal rod and a glass rod at one end for 1 minute. She touches the other ends. She will find: 1

**(a)** no change **(b)** glass rod hotter **(c)** both same **(d)** metal rod hotter

**9.** Why does a puddle disappear faster than a lake in the same sun? 1

**(a)** lake is salty
**(b)** smaller volume and large surface to depth ratio
**(c)** puddle is hotter
**(d)** lake has no water

**10.** A wheel of circumference 2.5 m makes 480 turns. Distance moved is: 1

**(a)** 480 m **(b)** 2400 m **(c)** 120 m **(d)** 1200 m

**11.** An iron cube and a wooden cube of the same size are placed in water. The iron sinks and the wood floats. What does this show? 1

**(a)** iron is soluble
**(b)** iron is heavier for the same size
**(c)** wood is heavy
**(d)** wood is magnetic

**12.** Which conclusion follows if the temperature of an object is not changing though heat is flowing in and out? 1

**(a)** object is hot
**(b)** no heat flow
**(c)** heat in equals heat out
**(d)** object is cold

**13.** **Assertion (A):** Evaporation takes place at all temperatures. 1
**Reason (R):** Boiling takes place only at a fixed temperature, about 100 degrees C.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**14.** **Assertion (A):** Standard units are needed to measure length. 1
**Reason (R):** Handspan and footstep differ from person to person.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

**15.** **Assertion (A):** Wood is an opaque material. 1
**Reason (R):** Frosted glass is a translucent material.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

## SECTION — B (Very Short Answer Type) 5 × 2 = 10

*Answer in about 20–30 words each.*

**16.** Why do we choose rubber for the handle of an electrician's tools? 2

**17.** Why do some thermometers use alcohol coloured red instead of mercury, especially at low temperatures? 2

**18.** How does a glass of cold drink 'sweat' even though the glass has no holes? 2

**19.** Why is the motion of a swing considered periodic but that of a falling leaf not? 2

**20.** How can the same material produce different objects? 2

## SECTION — C (Short Answer Type) — 5 × 3 = 15

*Answer in about 40–60 words each.*

**21.** Why do hill stations often receive more rainfall? 3

**22.** A drawing is made to a scale of 1 cm for 5 m. A building is 40 m long. What is its length in the drawing? 3

**23.** Why are bells and utensils made of metals? 3

**24.** What are the possible errors in using a thermometer to read the temperature of a liquid? 3

OR

Describe two ways in which forests help the water cycle. 3

**25.** A thread wound 25 times round a pencil of uniform thickness measures 8 cm in length of the coil. Find the thickness of the thread. 3

OR

Describe an experiment to show that air is a poor conductor of heat using materials of your choice. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**26.** **Case study.** Mansi's mother in Nagpur has a steel kadhai with a wooden handle, a glass bowl, a cloth bag and a plastic bucket. She showed that the glass bowl lets light through clearly, tracing paper lets some light through, and a wooden board lets none. 5

(i) Name the material of the kadhai.

(ii) Which material from the story is transparent?

(iii) Which is opaque?

(iv) Why is the handle of a kadhai made of wood and the kadhai of steel?

**27.** Compare traditional methods (matka, quilt, mud houses) with modern ones (fridge, thermos, thermocol) for controlling temperature. 5

OR

Analyse: 'Heating curve of ice' with data: 0 min -10 °C; 4 min 0 °C; 14 min 0 °C; 22 min 100 °C; 60 min 100 °C. 5

## END OF QUESTION PAPER — Total: 50 Marks